Final Project

CS520 - Graph Theory with Applications to Computer Science
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Introduction

“Maternal mortality (MMR) is widely acknowledged as a general indicator of the overall health of a population, of the status of women in society, and of the functioning of the health system”[1]. Motivated by a personal interest in women's health outcomes, I am seeking to apply graph algorithms to two public health metrics, Maternal Mortality Rate (MMR) and Infant Mortality Rate (IMR), to characterize geographic disparity in these two metrics across the contiguous 48 states. Combined, these two metrics provide a picture of the overall health of a population.

The states form a natural network when connected by geographic adjacency, weighted by the absolute difference in metric rates between neighboring states. Examining the raw data for each metric makes clear that metric disparity is not uniformly distributed: even neighboring states can exhibit drastically different rates, suggesting underlying policy, economic, or demographic factors.

Through this analysis, I aim to surface the following structural features of these metric networks:

- Regional clusters of similar metric rates
- Sharpest rate differences between neighboring states
- Broad geographic corridors of relatively high and low rates
- Cross-metric similarity

By exposing patterns of healthy and unhealthy regions, I hope to offer public health professionals a strategic starting point for further investigation into the root causes driving metric disparity across the US, which might ultimately allow for future policy recommendations to more effectively improve health outcomes for these metrics.

Problem Statement

$G = (V, E)$ is undirected, where V is the states in the contiguous 48 states with available data (Vermont is excluded as data was suppressed), and E is the geographic adjacency per state (two states share an edge if they share a border). For example if $v = OR$, then $E(v) = \{\{OR, CA\}, \{OR, WA\}, \{OR, NV\}, \{OR, ID\}\}$. For each metric $m \in \{MMR, IMR\}$, define a weighted graph $Gm = (V, E, w_m)$ where the edge weight $w_m(v_i, v_j) = |r_m(v_i) - r_m(v_j)|$, with $r_m(v) > 0$, the metric rate for the state v. I aim to derive this useful data by solving the following problems:

- **Find the minimum spanning tree T_m.** This surfaces regional clusters and sharpest rate differences between neighboring states. Cut edges, edges that are removed from T_m to partition T_m into subtrees and determined by the removal of edges with the heaviest weights.
- **Compute all pairs shortest path distance matrix D_m.** This surfaces broad geographic corridors of states with similar rates, which are long paths in D_m with low cumulative rate difference. Average metric rate per state in each path is the mean of $r_m(v)$ over all states in the path, and distinguishes between high and low rate corridors within D_m.
- **Compare T_{MMR} and T_{IMR} via Jaccard similarity[2], and D_{MMR} and D_{IMR} via Pearson correlation[3].** This describes metric similarity in regional clusters and metric correlation across all state pairs.

Methodology

MST Construction and cut edge identification
Kruskal's algorithm sorts all edges by weight and iterates through them from lightest to heaviest. For each edge, it checks if the vertices belong in the same connected component. If they do, it discards the edge to prevent a cycle, and if not, then it safely adds the edge to bridge the separate components. The result is the minimum spanning tree. The resulting MST captures the connections that are most similar by rate across the network. Edges removed form T_m, call these cut edges, are those with the highest weight, and represent the the sharpest rate transitions between neighboring states. Removing these edges partitions T_m into subtrees of states with similar rates, the regional clusters.

APSP distance matrix and corridor analysis

Floyd-Warshall's algorithm numbers vertices 1 to V and uses a triple-nested loop to find the shortest path between every pair of vertices u and v. In each phase r, it checks if routing a path from u to v through vertex r as an intermediate step is shorter than the best path found so far that doesn't use r. The resulting matrix stores the distance between a state pair, which I define as the cumulative metric rate increase along its shortest path. The algorithm also yields the paths along each state pair's shortest path. After computing distance per shortest path length, I isolate geographic corridors, which are paths in the matrix with low cumulative metric rate increase. I identify corridors with good and poor rates by comparing average metric rate among geographic corridors.

Cross-metric comparison

To determine if regional clusters hold across metrics, I use the Jaccard index. The Jaccard index is computed by the size of the intersection of two sets divided by the size of the union10. In this case, the sets are the edge sets of T_{MMR} and T_{IMR}. The resulting ratio is an indication of how many edges in each MST are shared. The closer the index to 1, the more similar the two sets. Since the graphs are weighted by the metric difference between states, if the regional clusters were similar across metrics, we might expect to see the same edges between state pairs in the MSTs. To determine if geographic corridors hold across metrics, I use Pearson correlation coefficient. The Pearson correlation coefficient measures the linear correlation between two sets of numeric data11. In this case, the two sets are the distance values, defined above, for each APSP matrix. The resulting coefficient is an indication of how correlated the MMR distance and IMR distance are for every state pair (1,081 pairs in total). In other words, it shows the degree to which the cumulative rate increase across a path between states is similar between metrics. A coefficient near 1 indicates that cumulative rate increases between state pairs are consistently similar across metrics and that the metric disparity between state pairs is similar across metrics on average.

Experimental Setup

Dataset

I used three datasets from the U.S. Center's for Disease Control's Wonder Database to compute MMR and IMR. CDC Wonder does not have a single query for MMR, so I had to compute it using number of pregnancy related deaths per state and number of live births per state. Specifically, I queried by underlying cause of death with parameters associated with pregnancy, childbirth, and puerperium[4] and live births per state[5]. CDC Wonder supports querying for IMR per state directly, so obtaining that data was a simple query[6]. Despite the wide date range used, data was suppressed for the state of Vermont, given the low number of deaths, in order to protect privacy. As such, Vermont is therefore omitted from this analysis.

Preprocessing

To construct each graph, I first constructed a state adjacency list as a dictionary in Python. I extracted the metric rate for every state and its neighbor, computed the absolute value difference for the edge weight, and then added that edge with its weight to a NetworkX graph.

Experimental design

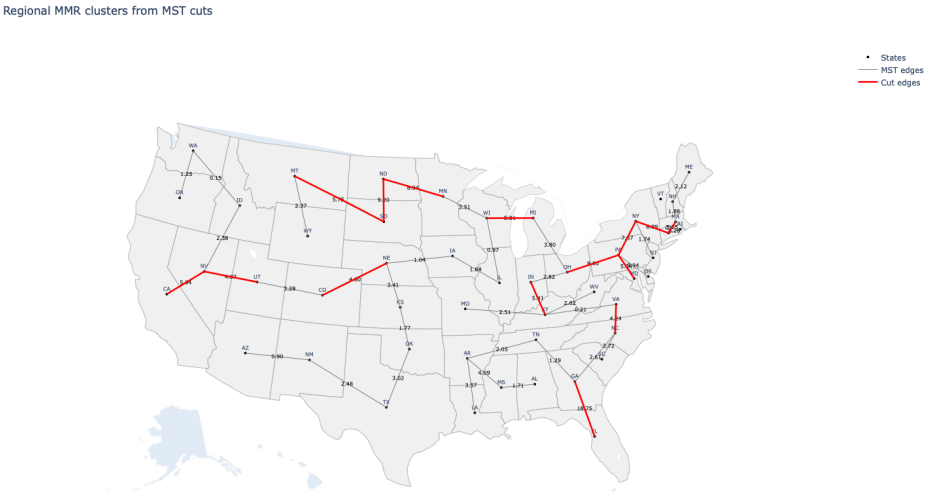
To find T_{MMR} and T_{IMR}, I ran Kruskal's algorithm on G_{MMR} and G_{IMR}. To construct D_{MMR} and D_{IMR}, I ran Floyd-Warshall's algorithm on the same graphs, storing path distance, hop count, distance per hop, states along each path, and average metric rate along each path. To assess cross-metric similarity, I computed Jaccard similarity on T_{MMR} and T_{IMR}, and Pearson correlation on D_{MMR} and D_{IMR}. Algorithms were implemented using NetworkX. Jaccard and Pearson computations were done using Pandas.

Parameters and configuration

With the exclusion of Vermont, and based on the state adjacency definition, each metric graph consists of 47 vertices and 104 edges. Given deaths caused by pregnancy, childbirth, and puerperium are low, MMR is computed as $(\text{number of maternal deaths}/\text{live births}) * 100000$[7]. Conversely, IMR is computed as $(\text{number of infant deaths}/1000 \text{ live births})$, where an infant is defined as a child between 0 and 11 months inclusive[8]. Each data set covers the year range 2018-2024. I used Jenks natural breaks optimization to determine MST cut edges in order to minimize variance within classes and maximize the variance between classes[9]. To select the geographic corridors with the smallest rate increase, I selected the top five paths with path length in the top 25% of path lengths, in order to ensure corridors span a meaningful number of states.

Results

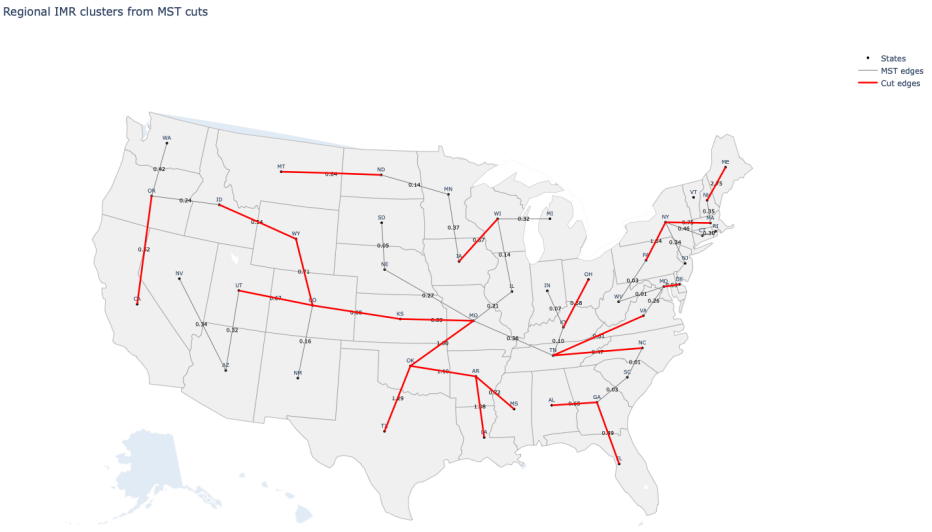
Regional clusters and states with sharpest rate difference



MST of G_{MMR}

Cutting the heaviest three Jenks classes yields 16 clusters: a Pacific Northwest cluster (OR, WA, ID, NV), a Northeast cluster (ME, NH, MA), a Southwest-to-Great-Lakes cluster (AZ, NM, TX, OK, KS, NE, IA, IL, WI, MN), and a Deep South cluster (LA, AR, MS, AL, TN, GA, NC, SC), along with other smaller clusters and 5 singleton outliers (CA, SD, ND, FL, MD). The average cluster edge weight is 2.20 vs. an average cut edge weight of 7.09. Cut edges are on average 3.22 times heavier than cluster edges, highlighting the sharpest rate transitions between neighboring states. The sharpest rate transition locations were:

`["FL", "GA"], ["CA", "NV"], ["CO", "NE"], ["NV", "UT"], ["CT", "MA"], ["CT", "NY"], ["NY", "PA"], ["MD", "PA"], ["PA", "OH"], ["NC", "VA"], ["MN", "ND"], ["SD", "MT"], ["SD", "ND"], ["WI", "MI"], ["IN", "KY"].`

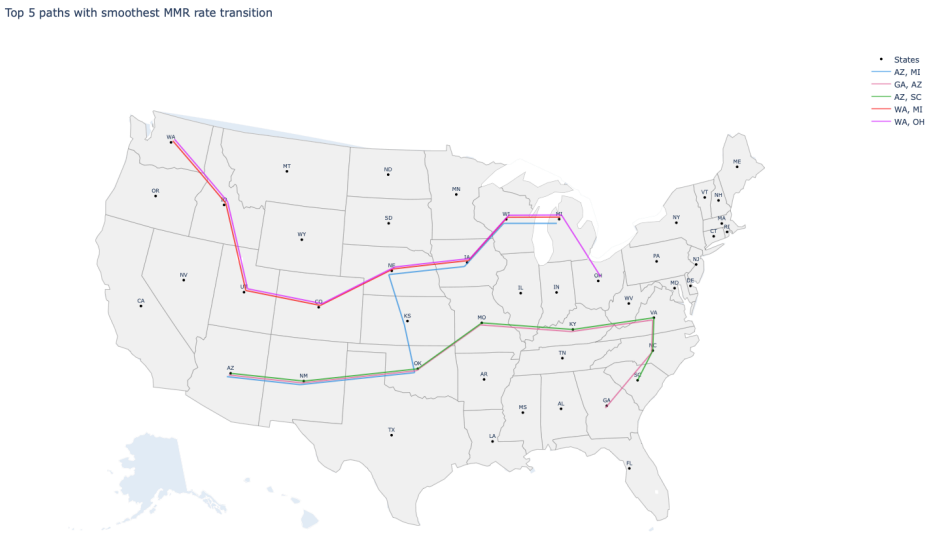


MST of G_{IMR}

Cutting the heaviest three Jenks classes yields 24 clusters: a Pacific Northwest cluster (OR, WA, ID), a Northeast cluster (NJ, NY, CT, RI), a Mid-Atlantic cluster (MD, WY, VA, PA), a Center cluster(SD, NE, MO, IL, TN, WI), a Southwest cluster (AZ, NV, UT), along with other smaller clusters and 11 singleton outliers (CA, WY, MT, TX, OK, AR, LA, MS, ME, AL, FL). The average cluster edge is weight is 0.22 vs. average cut edge weight of 0.88. Cut edges are on average 4 times heavier than cluster edges, highlighting the sharpest rate transitions between neighboring states. The sharpest rate transition locations were:

`["AL", "GA"], ["FL", "GA"], ["MS", "AR"], ["TN", "NC"], ["TN", "VA"], ["AR", "LA"], ["AR", "OK"], ["MO", "KS"], ["MO", "OK"], ["OK", "TX"], ["CA", "OR"], ["CO", "UT"], ["CO", "KS"], ["CO", "WY"], ["WY", "ID"], ["MA", "NY"], ["NY", "PA"], ["DE", "MD"], ["IA", "WI"], ["MT", "ND"], ["KY", "OH"], ["NH", "ME"]`

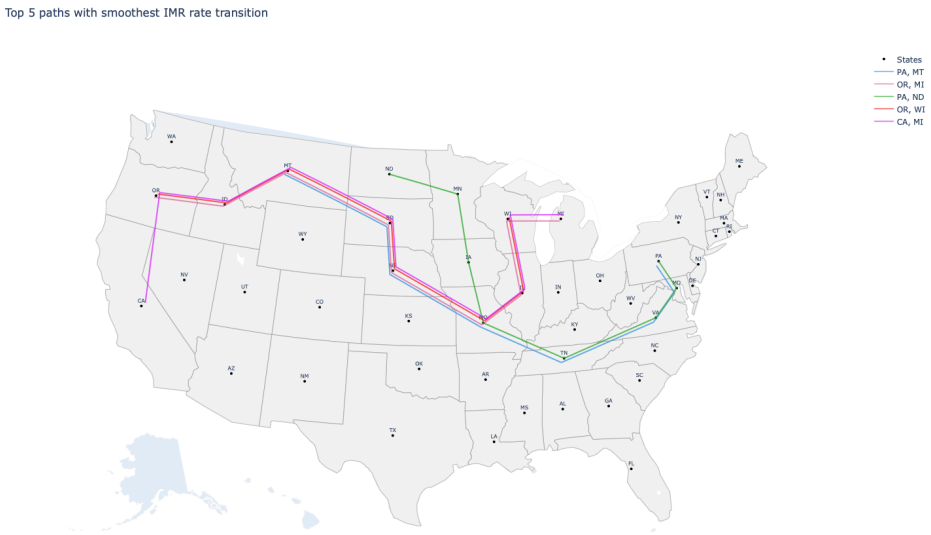
Geographic corridors of high and low rates



APSP on G_{MMR}

Sorting the rows of D_{MMR} by distance per hop ascending, then filtering to paths in the top 25% by hop count, yields the five geographic corridors with the smallest cumulative rate variation per hop. As the average distance per hop for selected geographic corridors is 1.82 times lighter than the average distance per hop of paths in D_{MMR} (3.32 vs. 6.07), the geographic corridors are substantially lighter than average as intended. Comparing the average state rate along each geographic corridor in D_{MMR} (33.56)to the average rate per state along path reveals three distinct rate categories among the corridors: two low, one near-average, and two high. Low rate geographic corridors are WA to OH (26.56 vs. 33.56) and WA to MI (25.68 vs. 33.56). The average rate corridor is AZ to MI (31.11 vs. 33.56). The high rate corridors are GA to AZ (40.79 vs. 33.56) and AZ to SC (40.12 vs. 33.56). Low rate corridors represent corridors with relatively health outcomes, and high rate corridors represent corridors of relatively unhealthy outcomes.

State 1	State 2	Distance	Hops	Distance / Hop	States in Path	Avg Rate Along Path
AZ	MI	21.23	7	3.03	[AZ, NM, OK, KS, NE, IA, WI, MI]	31.11
GA	AZ	22.26	7	3.18	[GA, NC, VA, KY, MO, OK, NM, AZ]	40.79
AZ	SC	22.37	7	3.19	[AZ, NM, OK, MO, KY, VA, NC, SC]	40.12
WA	MI	25.15	7	3.59	[WA, ID, UT, CO, NE, IA, WI, MI]	25.68
WA	OH	28.94	8	3.61	[WA, ID, UT, CO, NE, IA, WI, MI, OH]	26.56



APSP on G_{IMR}

Applying the same filtering and sorting as with D_{MMR} yields the five geographic corridors with the smallest cumulative rate variation per hop. As the average distance per hop for selected geographic corridors is 1.73 times lighter than the average distance per hop of paths in D_{IMR} (0.393 vs. 0.679), the geographic corridors are also substantially lighter than average for this metric. Unlike MMR, IMR rates along the selected corridors cluster tightly (5.52–5.99 within a 0.47 range), suggesting IMR does not exhibit the same good vs. poor corridor distinction observed for MMR for this approach.

State 1	State 2	Distance	Hops	Distance / Hop	States in Path	Avg Rate Along Path
MT	PA	2.42	7	0.346	[PA, MD, VA, TN, MO, NE, SD, MT]	5.99
MI	OR	3.19	8	0.399	[OR, ID, MT, SD, NE, MO, IL, WI, MI]	5.68
ND	PA	2.80	7	0.400	[PA, MD, VA, TN, MO, IA, MN, ND]	5.54
OR	WI	2.87	7	0.410	[OR, ID, MT, SD, NE, MO, IL, WI]	5.63
CA	MI	3.71	9	0.412	[CA, OR, ID, MT, SD, NE, MO, IL, WI, MI]	5.52

Cross metric similarity

The Jaccard index for the edge sets of T_{MMR} and T_{IMR} is 0.23, which indicates a low degree of similarity between MSTs. In other words, the regional clusters between metric rates do not overlap substantially, and the sharp-transition borders differ across metrics. This is supported by the map plot visualization which reveals relatively small number of overlapping edges between MSTs and the difference in cut edges between metrics. Conversely, the Pearson coefficient between state-pair distances of D_{MMR} and D_{IMR} is 0.79, which points to a strong correlation between state-pair distances across metrics. This means that disparity between state-pairs is similar for MMR and IMR on average. So while the regional structures and locations of sharp rate change across borders differs across metrics, the underlying disparity is strongly correlated across rates. This might suggest that socioeconomic, demographic, and healthcare access factors impact outcomes at a broad regional scale.

Conclusion

Ultimately, the analysis yields several actionable pathways for future public health investigation. A logical next step would be to expand the MST analysis to identify healthy and unhealthy clusters as I did with the geographic corridors. By identifying these clusters, researchers could investigate the root causes driving these opposing health outcomes, and then use the mechanisms of healthy clusters to inform policy interventions in unhealthy ones. Given the strong correlation between metrics, it is reasonable to expect that policy actions targeting one metric will yield a secondary benefit across the other. While the distinct regional differences imply that policy recommendations must also be localized, the subtrees and cut edge lists offer concrete starting points. To prioritize these efforts, I would start with the area where the rate transition is sharpest ((FL, AL) for MMR, and (ME, NH) for IMR). The findings suggest that addressing national health disparities requires an approach that balances both broad regional strategies and highly localized efforts; an insight which is surely no surprise to those in the public health sector.

References

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